Time Series - Estimation and model selection

Advanced Statistics and Alternative Data Types

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Introduction

- Consider a stationary, causal and invertible ARMA(p, q) model:

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = \epsilon_t + \theta_1 \epsilon_{t+1} + \cdots + \theta_q \epsilon_{t+q}, \quad \epsilon_t \sim \mathrm{WN}(0, \sigma^2).$$

- Today we will discuss two related issues:
  1. How to estimate $\phi = (\phi_1, \ldots, \phi_p)$, $\theta = (\theta_1, \ldots, \theta_1)$ and σ^2, given p, q
  2. How to choose p, q
- Today we will let $\vartheta = (\phi, \theta, \sigma^2)$ and $\beta = (\phi, \theta)$
- For AR models, OLS can be used to estimate the parameters
- For MA and ARMA models ($p > 0$), we typically use maximum likelihood

Maximum likelihood estimation

- Suppose we observe data, X, drawn from a distribution which has pdf $f_\vartheta(x) = f(X; \vartheta)$
- Given ϑ, f_ϑ, the pdf, is a function of X.
- We can also define the likelihood, $L_X(\vartheta) = f(X; \vartheta)$. Here we treat the data X as fixed; the likelihood is a function of ϑ.
- Maximum likelihood estimation chooses as our estimate of ϑ the value which maximses the likelihood over some parameter space Θ:

$$\hat{\vartheta} = \text{argmax}_{\vartheta \in \Theta} L_X(\vartheta).$$

- Such estimates often have good properties

Maximum likelihood estimation for (Gaussian) ARMA models

- To estimate the parameters θ of a causal and invertible ARMA process $(X_t)_{t\in\mathbb{Z}}$ we often proceed as if the process were Gaussian with mean zero and covariance function $\kappa(s,t) = \mathbb{E}[X_s X_t]$.
- Let $X^n = (X_1, \ldots, X_n)'$ and $\Gamma_n = \mathrm{Var}(X^n)$
  - Note that the variance matrix $\Gamma_n = \Gamma_n(\vartheta)$; it is a function of ϑ.
- The likelihood function in this case takes the form:

$$L_{X^n}(\vartheta) = (2\pi)^{-n/2} \left(\det \Gamma_n(\vartheta)\right)^{-1/2} \exp\left(-\frac{1}{2}(X^n)'\Gamma_n(\vartheta)^{-1}X^n\right).$$

- Our estimate of ϑ is the value which maximises $L_{X^n}(\vartheta)$ over a parameter space we choose. Typically this constrains the estimate to satisfy some required properties (i.e. that the parameters would define a causal, invertible ARMA model)

Maximum likelihood estimation for (Gaussian) ARMA models

- In practice, we maximise the likelihood function numerically
- Calculation of $\det \Gamma_n(\vartheta)$ and $\Gamma_n(\vartheta)^{-1}$ is computationally expensive for large n
- The procedure can be made more efficient by using the innovations algorithm
- Let $\theta_{n,j}$ and $\nu_n = \sigma^2 r_n$ be the coefficients and mean squared errors calculated by the innovations algorithm applied to the covariance function $\kappa_\vartheta(s,t)$
  - i.e. as though ϑ was the true parameter
- Define the matrices

$$C = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ \theta_{1,1} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \theta_{n-1,n-1} & \theta_{n-1,n-2} & \cdots & 1 \end{bmatrix}, \qquad D = \sigma^2 \begin{bmatrix} r_0 & 0 & \cdots & 0 \\ 0 & r_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & r_{n-1} \end{bmatrix}.$$

Maximum likelihood estimation for (Gaussian) ARMA models

- By the definition of these coefficients, we have

$$\hat{X}^n = (\hat{X}_1, \ldots, \hat{X}_n)' = (C - I_n)(X^n - \hat{X}^n) \quad \Longrightarrow \quad X^n = C(X^n - \hat{X}^n)$$

- D is the covariance matrix of $(\hat{X}_n - X_n)$ [see the notes] so $\Gamma_n = \mathrm{Var}(X^n) = CDC'$ and therefore

$$\det \Gamma_n = \det(C)^2 \det(D) = \sigma^{2n} \prod_{t=1}^{n} r_{t-1},$$

$$\begin{aligned}
(X^n)' \Gamma_n^{-1} X^n &= (X^n - \hat{X}^n)' C' \Gamma_n^{-1} C (X^n - \hat{X}^n) \\
&= (X^n - \hat{X}^n)' C' (C')^{-1} D^{-1} C^{-1} C (X^n - \hat{X}^n) \\
&= (X^n - \hat{X}^n)' D^{-1} (X^n - \hat{X}^n) \\
&= \sigma^{-2} \sum_{t=1}^{n} \frac{(X_t - \hat{X}_t)^2}{r_{t-1}}.
\end{aligned}$$

Maximum likelihood estimation for (Gaussian) ARMA models

- Plugging these expressions into our likelihood equation we obtain

$$L_{X^n}(\vartheta) = (2\pi\sigma^2)^{-n/2} \left(\prod_{i=0}^{n-1} r_i \right)^{-1/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{t=1}^{n} \frac{(X_t - \hat{X}_t)^2}{r_{t-1}} \right).$$

- We can maximise this equation to obtain an estimate of ϑ.
- Even if $(X_t)_{t\in\mathbb{Z}}$ is not Gaussian, we can estimate the parameters in this way:
  - The expression above will be smaller the closer the best linear predictor $\hat{X}_t$ is to the true X_t
  - The best linear predictor depends only on $\kappa_\vartheta(s,t)$
  - Heuristically: the closer $\kappa_\vartheta(s,t)$ is to the true autocovariance function, the closer $\hat{X}_t$ is to X_t

Maximum likelihood estimation for (Gaussian) ARMA models

- For $\hat{X}_t$ and r_{t-1} the outputs of the innovation algorithm applied to the covariance function of a transformed version of the process X_t under ϕ, θ [see notes for details],[1] define

$$S(\phi, \theta) = \sum_{t=1}^{n} \frac{(X_t - \hat{X}_t)^2}{r_{t-1}}$$

- Let the maximisers of $L_{X^n}(\vartheta)$ be $\hat{\vartheta} = (\hat{\phi}, \hat{\theta}, \hat{\sigma}^2)$
- $\hat{\phi}$ and $\hat{\theta}$ are the values which minimise

$$l(\phi, \theta) = \log\left(\frac{1}{n} S(\phi, \theta)\right) + \frac{1}{n} \sum_{i=1}^{n} \log r_{n-i}$$

- $\hat{\sigma}^2$ satisfies

$$\hat{\sigma}^2 = S(\hat{\phi}, \hat{\theta})/n.$$

[1] These $\hat{X}_t$ and r_{t-1} do *not* depend on σ^2

Maximum likelihood estimation for (Gaussian) ARMA models

Theorem

Suppose that $(X_t)_{t\in\mathbb{Z}}$ *is a causal and invertible ARMA*(p,q) *process where* Φ *and* Θ *have no common factors and* $\epsilon_t \sim IID(0,\sigma^2)$.
Let C *be the set of values of* $\beta=(\phi,\theta)$ *such that if* Φ' *and* Θ' *are the implied lag polynomials,* $\Phi'(z)\Theta'(z)\neq 0$ *for* $|z|\leq 1$, $\phi_p\neq 0$, $\theta_q\neq 0$ *and* Φ' *and* Θ' *have no common factors.*
Then, if $\hat{\beta}=\operatorname{argmin}_{\beta\in C} l(\phi,\theta)$,

$$\sqrt{n}(\hat{\beta}-\beta)\rightsquigarrow \mathcal{N}(0,\Sigma(\beta)),\quad \Sigma(\beta)=\sigma^2\begin{bmatrix}\mathbb{E}[U^t(U^t)'] & \mathbb{E}[U^t(V^t)']\\ \mathbb{E}[V^t(U^t)'] & \mathbb{E}[V^t(V^t)']\end{bmatrix}^{-1},$$

where $U^t=(U_t,\ldots,U_{t+1-p})'$, $V_t=(V_t,\ldots,V_{t+1-q})'$ *are the autoregressive processes*

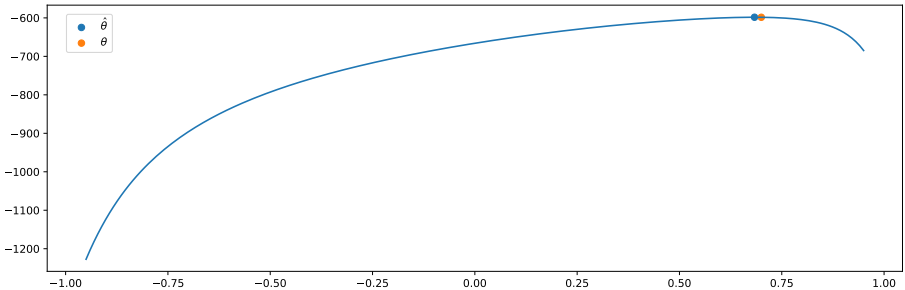
$$\Phi(L)U_t=\epsilon_t,\qquad \Theta(L)V_t=\epsilon_t.$$

If $p=0$, *then* $\Sigma(\beta)=\sigma^2\mathbb{E}[V^t(V^t)']^{-1}$ *and if* $q=0$ *then* $\Sigma(\beta)=\sigma^2\mathbb{E}[U^t(U^t)']^{-1}$.

Example: MA(1)

- We draw 400 samples from a MA(1) with true parameter $\theta = 0.7$ and i.i.d. $t(8)$ errors.
- We then estimated the model by maximum likelihood Code

Figure 1: $\log L_{X^n}(\theta)$



Example: MA(1)

Figure 2: Statsmodels output

```
>>> res.summary()
<class 'statsmodels.iolib.summary.Summary'>
"""
                               SARIMAX Results
==============================================================================
Dep. Variable:                      y   No. Observations:                  400
Model:                 ARIMA(0, 0, 1)   Log Likelihood                -598.331
Date:                Sun, 30 Oct 2022   AIC                           1200.662
Time:                        14:46:45   BIC                           1208.645
Sample:                             0   HQIC                          1203.823
                                  - 400   Scale                            1.164
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ma.L1          0.6834      0.040     17.096      0.000       0.605       0.762
===================================================================================
Ljung-Box (L1) (Q):                   0.87   Jarque-Bera (JB):                 7.42
Prob(Q):                              0.35   Prob(JB):                         0.02
Heteroskedasticity (H):               0.90   Skew:                             0.03
Prob(H) (two-sided):                  0.54   Kurtosis:                         3.66
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
"""
```

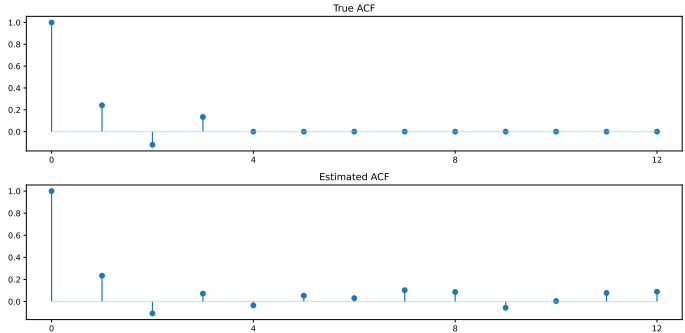
Order selection

- Thus far we have discussed estimation given that we know p, q
- In reality we do not known p, q and therefore need also to establish p, q from our data
- We will discuss two points:
  1. The identification of pure AR and MA process from the ACF and PACF
  2. Identification of p, q via information criteria

Pure AR & MA models and the ACF & PACF

- Let ρ denote the autocorrelation function (ACF)
- The ACF is useful to identify the order of a pure MA model. A MA(q) model has $\rho(j) = 0$ for all $|j| > q$:

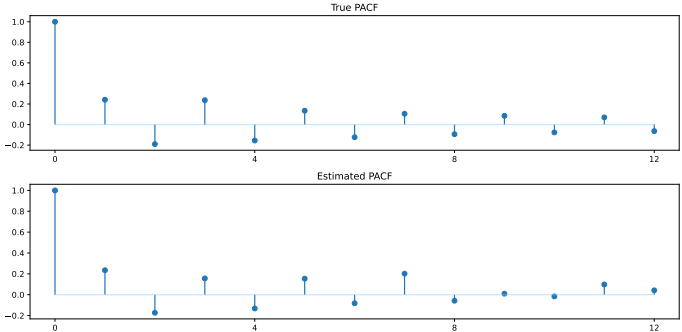
Figure 3: ACF and estimated ACF of a MA(3)



Pure AR & MA models and the ACF & PACF

- Let α denote the partial autocorrelation function (PACF)
- A MA(q) model has exponentially declining α

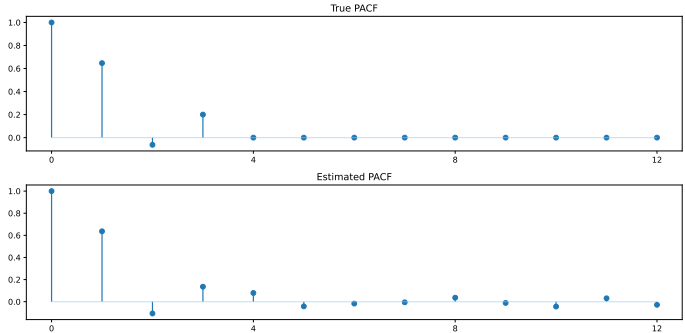
Figure 4: PACF and estimated PACF of a MA(3)



Pure AR & MA models and the ACF & PACF

- Let α denote the partial autocorrelation function (PACF)
- The PACF is useful to identify the order of a pure AR model. A AR(p) model has $\alpha(j) = 0$ for all $|j| > p$:

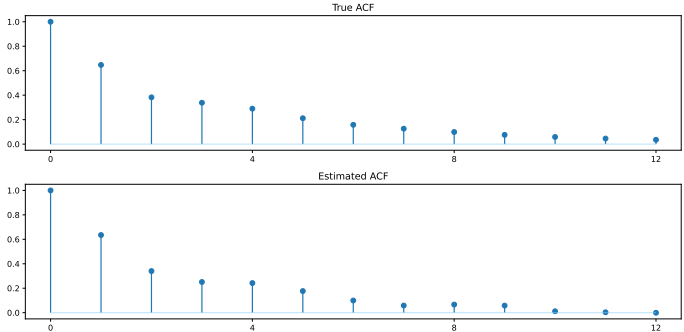
Figure 5: PACF and estimated PACF of a AR(3)



Pure AR & MA models and the ACF & PACF

- Let ρ denote the autocorrelation function (ACF)
- A AR(p) model has exponentially declining ρ

Figure 6: ACF and estimated ACF of a AR(3)



Pure AR & MA models and the ACF & PACF

- $\rho(j) = 0$ for all $j > q$ and α exponentially declining $\implies$ MA(q)
- $\alpha(j) = 0$ for all $j > p$ and ρ exponentially declining $\implies$ AR(p)
- Both ρ and α exponentially declining $\implies$ ARMA(p, q)

Information criteria

- The estimated ACF / PACF may indicate a pure AR or MA model but the order may not be clear
- The estimated ACF / PACF may indicate an ARMA model
- In either of these cases we still need to determine the value of p, q or both
- The idea of information criteria is regularisation
- Just as in linear regression models, we can typically achieve a better in-sample fit by including more parameters (i.e. large p and/or q)
- This comes with the risk of overfitting
- We avoid this by adding a penalty function to the likelihood, which penalises large models

Information criteria

- The resulting objective functions are called information criteria

$$AIC(\phi, \theta) = -2 \log L(\phi, \theta, S(\phi, \theta)/n) + 2(p + q + 1)$$
$$AICC(\phi, \theta) = -2 \log L(\phi, \theta, S(\phi, \theta)/n) + \frac{2(p + q + 1)n}{n - p - q - 2}$$
$$BIC(\phi, \theta) = -2 \log L(\phi, \theta, S(\phi, \theta)/n) + (p + q + 1) \log n.$$

- The AIC and AICC have very similar forms, but the AICC penalises large order (higher p, q) models more strongly.
- The BIC can be shown to have a consistency property: if $\hat{p}_n$ and $\hat{q}_n$ are the chosen orders and $(X_t)_{t \in \mathbb{Z}}$ is an ARMA(p, q) process then as $n \to \infty$, $\hat{p}_n \xrightarrow{P} p$ and $\hat{q}_n \xrightarrow{P} q$
- The AIC and AICC do not have this consistency property but they do have an efficiency property which the BIC does not share. See section 5.5 of [1] for further detail and references.

Information criteria

- Information criteria are used as follows: p, q and the corresponding $\phi_1, \ldots, \phi_p$ and $\theta_1, \ldots, \theta_q$ are chosen to minimise $IC(\phi, \theta)$ where IC is one of $AIC, AICC, BIC$
- After fitting the model you should always check that the implied residuals resemble white noise. See section 5.3 of [1] for further details

References I

[1] P. J. Brockwell and R. A. Davis. *Introduction to Time Series and Forecasting*. Springer Texts in Statistics. Springer New York, third edition, 2016.

Code for MA(1) estimation (1/2)

```
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm

np.random.seed(845723)
n = 400
burnin = 400

X = sm.tsa.ArmaProcess(ar = [1.0], ma = [1.0, 0.7]).generate_sample(n + burnin,
        distrvs = lambda size: np.random.standard_t(8, size)
        )[burnin:]
mod = sm.tsa.ARIMA(X, order = (0,0, 1), trend="n", concentrate_scale=True)
res = mod.fit()
res.summary()
```

◂ Results

Code for MA(1) estimation (2/2)

```
thetahat = res.params[0]
ll = mod.loglike(thetahat)
thetas = np.linspace(-0.95, 0.95, 200)
lls = [mod.loglike(theta) for theta in thetas]


fig, ax = plt.subplots(figsize = (12, 4))
ax.plot(thetas, lls)
ax.scatter([thetahat], [ll], label = r"$\hat\theta$")
ax.scatter([0.7], [mod.loglike(0.7)], label = r"$\theta$", color = "tab:orange")
fig.legend(loc = "upper left", bbox_to_anchor=(0.075, 0.95))
fig.tight_layout()
plt.savefig("output/ma1_loglik.pdf")
```

◂ Results